

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering – ASSESSMENT TOOL 1 – MINI APPLICATION DEVELOPMENT (CHATBOT / SUMMARIZER / TRANSLATOR)

Course Code:	CSA6503	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4, BL5)	Assessment:	Assessment Tool 1 - Mini Application Development (Chatbot / Summarizer / Translator)
Weightage:	20% of CIA	Total Marks:	2 * 50 = 100
CO4 Statement:	Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications. (BL3, BL4, BL5)		
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Section / Batch:	Slot D / 2024	Date:	24/08/2026

SECTION A – MINI APPLICATION DEVELOPMENT (CHATBOT / SUMMARIZER / TRANSLATOR)

AI TRANSLATOR APPLICATIONS

1. Campus Multilingual Translator – Develop an AI-powered translator application that converts admission notices, forms, and general campus information between English and regional/foreign languages. Provide a user-friendly interface and demonstrate the translator using at least 15 test phrases.

1. Campus Multilingual Translator – Project Title

AI-Powered Campus Multilingual Translator

Introduction

The Campus Multilingual Translator is an AI-based application designed to help students, parents, and international visitors understand admission notices, forms, and general campus information in their preferred language. Instead of relying on a staff member or manually searching for translations, users can type or paste text and instantly receive an accurate translation through a simple conversational interface.

The translator can convert text between English and regional/foreign languages such as Tamil, Hindi, Telugu, and French, covering content related to admissions, fees, hostel information, academic notices, and general campus announcements.

Objectives

- Provide instant and accurate translation of campus-related text.
- Reduce dependency on manual translation by staff.
- Help non-native speakers understand admission and academic information.
- Support multiple regional and foreign languages.
- Preserve the original meaning and context during translation.
- Improve accessibility of college communication for all users.
- Provide a simple and user-friendly interface.

Main Features

Feature	Description
Text Translation	Converts typed or pasted text between supported languages
Language Detection	Automatically identifies the source language of the input
Multi-Language Support	Supports English, Tamil, Hindi, Telugu, and French

Admission Notice Translation	Translates admission-related notices and circulars
Form Translation	Translates application forms and instructions
Fee & Scholarship Translation	Explains fee structures and scholarships in the target language
Pronunciation Guide	Provides phonetic pronunciation for translated text
Copy & Export	Allows users to copy or export translated text
History Log	Keeps a record of recent translations
Chat Interface	Allows users to interact naturally with the translator

Example Conversation

User: Translate: 'Admission forms must be submitted before the deadline.' into Tamil.

Translator: விண்ணப்பப் படிவங்களை கடைசி தேதிக்கு முன் சமர்ப்பிக்க வேண்டும்.

User: Translate: 'Hostel fees include accommodation and meals.' into Hindi.

Translator: छात्रावास शुल्क में आवास और भोजन शामिल है।

User: What language is 'Bienvenue au collège' written in?

Translator: This text is written in French and translates to 'Welcome to the college' in English.

15 Test Queries

You can demonstrate these during your project presentation:

1. Translate 'Welcome to our college' into Tamil.
2. Translate 'What is the last date for admission?' into Hindi.
3. Translate 'Please submit your transfer certificate' into Telugu.
4. Translate 'Scholarships are available for meritorious students' into French.
5. Detect the language of 'Nalla kaalai vanakkam'.
6. Translate 'Hostel accommodation is limited' into Hindi.
7. Translate 'The entrance exam will be held in June' into Tamil.
8. Translate 'Fees can be paid online' into Telugu.
9. Convert 'Contact the admission office for details' into French.
10. Translate a full admission notice paragraph into Tamil.
11. Translate 'Documents required: mark sheet, ID proof, photograph' into Hindi.
12. Provide the pronunciation of the Tamil translation of 'Good morning'.
13. Translate 'Classes begin from the first week of July' into Telugu.
14. Translate a scholarship notice into French.
15. Translate 'Thank you for visiting our college' into Hindi.

2. Academic Document Translator – Project Title

AI-Based Academic Document Translator

Introduction

The Academic Document Translator is an AI-based application that helps students understand academic materials such as notices, question papers, syllabi, and study material in their preferred language. It acts as a virtual language assistant and can translate content related to subjects, examinations, timetables, and academic announcements.

Students can paste or upload academic text into the translator and receive a translation without manually searching for meanings or relying on external help, making academic content more accessible to students from different linguistic backgrounds.

Objectives

- Provide quick translation of academic content for students.
- Help students understand notices and instructions in their native language.
- Translate examination and assignment-related information.
- Support translation of academic schedules and timetables.
- Reduce miscommunication caused by language barriers.
- Provide accurate, context-aware translations.
- Create an easy-to-use academic translation system.

Main Features

Feature	Description
Subject Notice Translation	Translates subject and syllabus-related notices
Assignment Translation	Translates assignment instructions and deadlines
Exam Notice Translation	Translates exam dates and instructions
Attendance Notice Translation	Translates attendance-related communication
Timetable Translation	Converts class schedule information into the target language
Academic Calendar Translation	Translates important academic dates
Document Upload	Allows translation of uploaded academic documents
Multi-Language Support	Supports English, Tamil, Hindi, and Telugu
Glossary Assistance	Explains academic terms during translation
Chat Interface	Provides conversational translation interaction

Example Conversation

Student: Translate: 'The semester examination begins next Monday.' into Tamil.

Translator: செமஸ்டர் தேர்வு அடுத்த திங்கள்கிழமை தொடங்குகிறது.

Student: Translate: 'Assignment submission deadline is Friday.' into Hindi.

Translator: असाइनमेंट जमा करने की अंतिम तिथि शुक्रवार है।

Student: Translate: 'Minimum attendance required is 75 percent.' into Telugu.

Translator: కనీస హాజరు అవసరం 75 శాతం.

15 Test Queries for Academic Translator

These are particularly useful because your assignment requires different types of translation queries.

Subject Queries

1. Translate 'What subjects do I have this semester?' into Tamil.
2. Translate the syllabus of Python programming into Hindi.
3. Translate 'How many subjects are there this semester?' into Telugu.

Assignment Queries

4. Translate 'When is my assignment submission deadline?' into Tamil.
5. Translate 'How can I submit my assignment?' into Hindi.
6. Translate a late-submission penalty notice into Telugu.

Examination Queries

7. Translate 'When are the semester examinations?' into Tamil.
8. Translate exam preparation instructions into Hindi.
9. Translate 'Where can I find the exam timetable?' into Telugu.

Attendance Queries

10. Translate 'What is the minimum attendance required?' into Tamil.
11. Translate an attendance shortage notice into Hindi.
12. Translate 'How can I check my attendance?' into Telugu.

Academic Schedule Queries

13. Translate 'When does the semester start?' into Tamil.
14. Translate the holiday list into Hindi.
15. Translate 'When are the internal examinations?' into Telugu.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Model/Technique Fit	30%	Correct, appropriate model/technique choice for the chosen application	Mostly appropriate choices made	Some inappropriate technique choices	Technique choice fundamentally flawed
Functional Completeness	25%	Application fully functional; solves a real use-case effectively	Mostly functional application	Partially functional application	Application is non-functional
Robustness & Reliability	20%	Handles edge cases; stable, reliable performance	Handles common cases well	Limited robustness to edge cases	Frequent failures during use
UI / Demo Polish	15%	Polished UI/demo with an engaging walkthrough	Clear demo, minor polish issues	Basic or rough demo	No usable demo presented
Design & Usage Documentation	10%	Comprehensive documentation of design & usage	Adequate documentation	Minimal documentation	No documentation provided

Marks Evaluation:

Question No.	Model/Technique Fit (30%)	Functional Completeness (25%)	Robustness & Reliability (20%)	UI / Demo Polish (15%)	Design & Usage Documentation (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						